

This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Part 1: Data

- ☐ This paper does not involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).
- ☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

The analyses use two main data sources: (1) EPA daily PM10 (AQI) data aggregated to weekly grid cells and saved as `data/epa-aqs/aqi-us.Rda` (public, processed); and (2) a GFR kidney-transplant dataset (`data/gfr/GFR2023Feb01.csv`) that is privately held and available by request. The README files (`data/README.md`, `application/README.md`) document preprocessing and formats used in the analyses.

Availability

- ☒ Data **are** publicly available.
- ☒ Data **cannot be made** publicly available.

Note: the repository contains a mix of data sources.

- The EPA AQI (PM10 series) raw CSVs are publicly available from EPA (see link below) and the processed AQI object `data/epa-aqs/aqi-us.Rda` is included in the repo.
- The GFR kidney-transplant dataset (`data/gfr/GFR2023Feb01.csv`) is private and must be requested from the data custodian (see “Non-publicly available data” below).

Publicly available data

- ☒ Data are available online at: https://aqs.epa.gov/aqsweb/airdata/download_files.html (EPA AirData, daily PM10 series 81102). The processed AQI dataset used in the analyses (`data/epa-aqs/aqi-us.Rda`) is included in this repository (see `data/README.md` and `application/README.md` for preprocessing details).
- ☐ Data are available as part of the paper’s supplementary material.
- ☐ Data are publicly available by request, following the process described here:
- ☐ Data are or will be made available through some other mechanism, described here:

Non-publicly available data

The GFR kidney-transplant dataset used in the application (`data/gfr/GFR2023Feb01.csv`) is subject to access restrictions and is not included in this repository. Data sharing is not possible through this repository due to custodial restrictions; interested researchers should request access from the data custodian as described in Liu et al. (2023). To aid reproducibility despite the restriction, (i) preprocessing steps and code to build per-subject lists (`Ly`, `Ltid`, `Lmi`) are provided in `application/gfr.R` and `data/README.md`, and (ii) simulation scripts that use estimated FPCs (e.g., `experiments/simu_gfr.R`) allow replication of the computational methods.

Description

File format(s)

- ☒ CSV or other plain text.

- ☒ Software-specific binary format (.Rda, Python pickle, etc.): .Rda file (`data/epa-aqs/aqi-us.Rda`)
- ☐ Standardized binary format (e.g., netCDF, HDF5, etc.):
- ☐ Other (please specify):

Data dictionary

- ☒ Provided by authors in the following file(s): `data/README.md`
- ☐ Data file(s) is(are) self-describing (e.g., netCDF files)
- ☒ Available at the following URL: https://aqs.epa.gov/aqsweb/airdata/download_files.html (for raw AQI CSVs)

Additional Information (optional)

Part 2: Code

Abstract

This repository contains R scripts implementing the OnlineFPCA algorithms (core code in `R/`), simulation pipelines (`experiments/`), application analysis scripts (`application/`), and helper scripts (e.g., `install_pkgs.R`). Analysis and plotting scripts produce the figures and tables used in the paper; detailed reproduction steps are in `README.md`.

Description

Code format(s)

- ☒ Script files
 - ☒ R
 - ☐ Python
 - ☐ Matlab
 - ☐ Other:
- ☐ Package
 - ☐ R
 - ☐ Python
 - ☐ MATLAB toolbox
 - ☐ Other:
- ☐ Reproducible report
 - ☐ R Markdown
 - ☐ Jupyter notebook
 - ☐ Other:
- ☐ Shell script
- ☐ Other (please specify):

Supporting software requirements

Version of primary software used R 4.5.0 (tested).

Libraries and dependencies used by the code Major R packages used for the algorithms (tested versions listed in `README.md`): `fda`, `fpca`, `fdapace`, `face`, `mgcv`, `gslnls`, `RSpectra`, `fastmatrix`, `ManifoldOptim`, `sm`, `Rcpp`, `RcppArmadillo`.

Other supporting packages for data manipulation, visualization, parallelization, etc.: `dplyr`, `tidyr`, `stringr`, `readr`, `readxl`, `lubridate`, `cowplot`, `foreach`, `doFuture`, `sf`, `spData`, `giscoR`, `Rdimtools`, `Metrics`, `rTensor`, `remotes`, `argparse`, etc.

See `install_pkgs.R` and `README.md` (in the root directory) for full list of packages and versions; `install_pkgs.R` automates installation on a fresh environment.

Supporting system/hardware requirements (optional)

No specific hardware requirements. Standard desktop or laptop machine with R installed should suffice for small-scale runs. Larger repeated simulation and some application analyses may benefit from access to a high-performance computing (HPC) cluster.

Parallelization used

- ☒ No parallel code used
- ☐ Multi-core parallelization on a single machine/node
 - Number of cores used:
- ☐ Multi-machine/multi-node parallelization
 - Number of nodes and cores used:

License

- ☒ MIT License (default)
- ☐ BSD
- ☐ GPL v3.0
- ☐ Creative Commons
- ☐ Other: (please specify)

Additional information (optional)

Part 3: Reproducibility workflow

Scope

The provided workflow reproduces:

- ☒ Any numbers provided in text in the paper
- ☒ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))
- ☒ All tables and figures in the paper
- ☐ Selected tables and figures in the paper, as explained and justified below:

Workflow

Location

The workflow is available:

- ☐ As part of the paper's supplementary material.
- ☒ In this Git repository:
 - `README.md`: the overall introduction and the summarized workflow to reproduce all tables and figures
 - `experiments/README.md`: steps to run the simulation studies
 - `application/README.md`: steps to run real data analysis
- ☐ Other (please specify):

Format(s)

- ☐ Single master code file
- ☐ Wrapper (shell) script(s)
- ☐ Self-contained R Markdown file, Jupyter notebook, or other literate programming approach
- ☒ Text file (e.g., a readme-style file) that documents workflow (`README.md`, `application/README.md`, `experiments/README.md`)

- ☐ Makefile
- ☐ Other (more detail in *Instructions* below)

Instructions

- Install required R packages: `Rscript install_pkgs.R` (see `install_pkgs.R` for the package list and `README.md` for tested versions).
- For simulation studies: run `Rscript experiments/fpca1d.R / fpca2d.R` (use `--seed` as needed) and then `Rscript experiments/analyze_fpca1d.R / analyze_fpca2d.R` to aggregate results and generate tables/figures.
- For application analyses:
 - Ensure required data is present: processed AQI data (`data/epa-aqs/aqi-us.Rda`) is included; raw EPA CSVs can be downloaded from EPA AirData if needed; the private GFR CSV is not included and must be placed in `data/gfr/` if available.
 - Then, run `Rscript application/aqi.R` and then `Rscript application/analyze_aqi.R` to reproduce AQI figures; run `Rscript application/gfr.R` and then `Rscript application/analyze_gfr.R` for GFR analyses.

See `README.md`, `experiments/README.md`, and `application/README.md` for more details and example commands.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:

- ☐ < 1 minute
- ☒ 1-10 minutes
- ☒ 10-60 minutes
- ☐ 1-8 hours
- ☐ > 8 hours
- ☐ Not feasible to run on a desktop machine, as described here:

Note: small demo scripts and single-seed runs complete quickly; full multi-seed simulation sweeps and some application runs may be long-running and are typically executed on an HPC cluster.

Additional information (optional)

Notes (optional)